

Most of the people in the world use the Celsius scale to measure temperature for day to day purpose. The Kelvin scale has been designed in such a way, it is not only an absolute temperature scale, but also 1°C change is equal to a 1K change. This makes the conversion from Celsius to absolute temperature scale (Kelvin scale) easy, just the addition or subtraction of a constant 273.15

But in United States they prefer to use the Fahrenheit scale. The problem is, converting Fahrenheit to absolute scale (Kelvin) is not easy.

To sort out this problem they use The Rankine scale. It named after the Glasgow University engineer and physicist **Rankine**, who proposed it in 1859. It is an absolute temperature scale, and has the property of having a 1°R change is equal to a 1°F change. Fahrenheit users who need to work with absolute temperature can be converted to Rankine by

R= F+ 459.67

1.6 Numerical Problems

Solved examples

1. How much will the temperature of 68°F be in Celsius and Kelvin?

Given :

Temperature in Fahrenheit = F = 68°F

Temperature in Celsius = C = ?

Temperature in Kelvin = K = ?

$$\frac{(F-32)}{9} = \frac{C}{5}$$

$$\frac{(68-32)}{9} = \frac{C}{5}$$

$$C = 5x\frac{36}{9} = 20°C$$

$$K = C + 273.15 = 20 + 273.15 = 293.15$$

Thus, the temperature in Celsius = 20ºC and in Kelvin = 293.15 K

2. At what temperature will its value be same in Celsius and in Fahrenheit?

Given : If the temperature in Celsius is C, then the temperature in Fahrenheit (F) will be same,

i.e. F = C. $\frac{(F-32)}{9} = \frac{C}{5}$

(or)

$$\frac{(C-32)}{9} = \frac{C}{5}$$

$$(C–32) \times 5 = C \times 9$$

$$5C – 160 = 9\,C$$

$$4\,C = – 160$$

$$C = F = – 40$$

The temperatures in Celsius and in Fahrenheit will be same at – 40

3. Convert the given temperature :

1) 45°C = ... °F 2) 20°C = ... °F

3) 68°F = ... °C 4) 185°F = ... °C

5) 0°C = ... K 6) -20°C = ... K

7) 100 K = ... °C 8) 272.15 K = ... °C

POINTS TO REMEMBER

1. The measurement of warmness or coldness of a substance is known as its temperature.
2. There are three units which are used to measure the temperature: Degree Celsius, Fahrenheit and Kelvin.
3. The SI unit of temperature is Kelvin (K).
4. In a thermometer, when liquid gets heat, it expands and when it is cooled down, it contracts. It is used to measure temperature.

VII Std Science Term-2 EM Unit 1.indd 9 20-07-2019 1.46.42 PM